

Materialized View Log Evolution

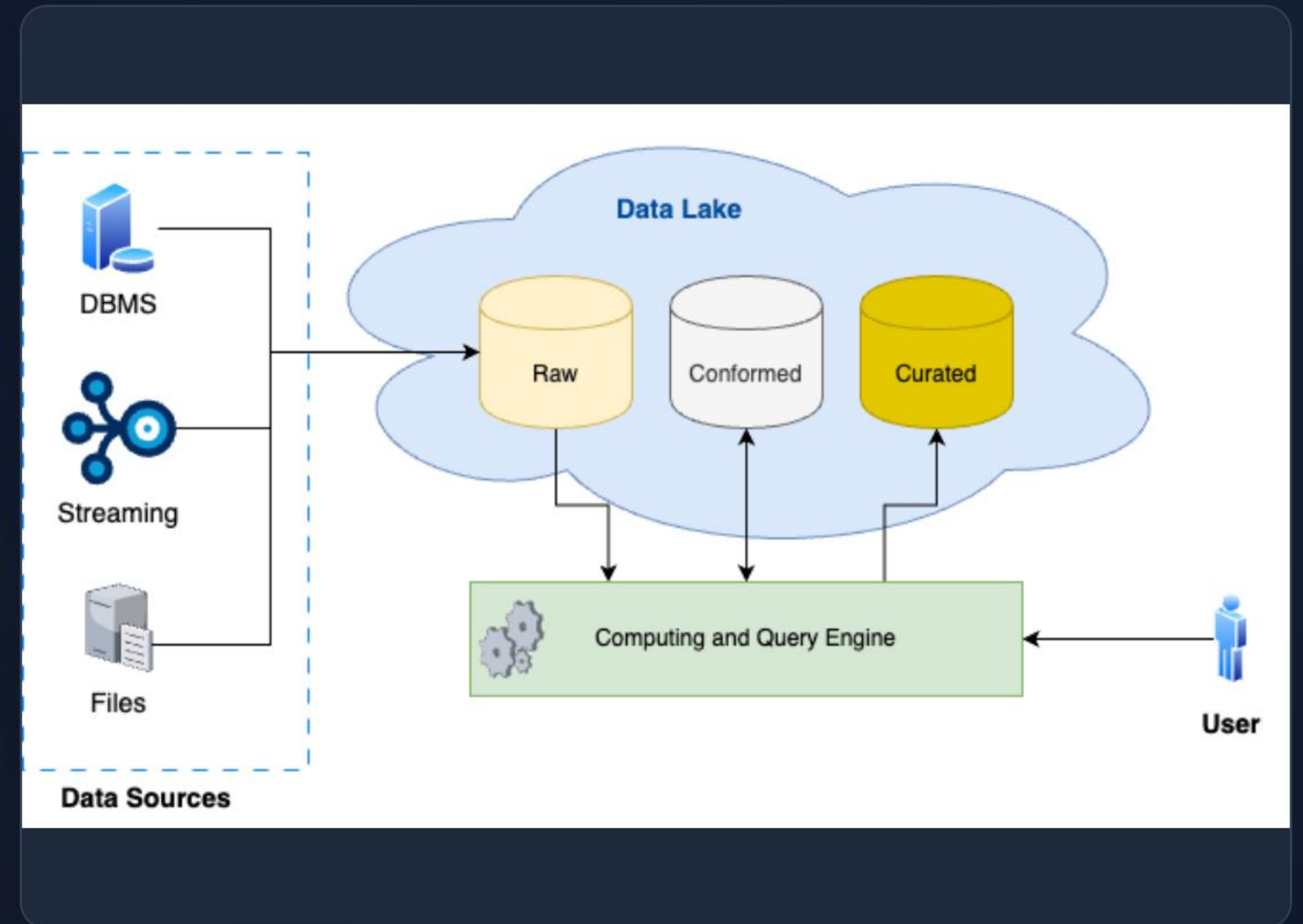
Ensuring FAST Refresh Stability During Schema Changes

The Challenge of Schema Evolution

Why altering a base table breaks the replication pipeline.

Understanding the Mismatch

- 🗄️ MV Logs track DML changes as independent database objects.
- 🔗 Adding a column to a table does not automatically update the log metadata.
- ⚠️ Missing columns in logs force Oracle to default to complete refreshes.



The Performance Impact

95%

Increased Latency

The Cost of Full Refresh

When FAST refresh fails, the database must scan the entire source table. For multi-terabyte tables, this leads to significant I/O overhead and prolonged synchronization windows.

Strategic Resolution Paths

Strategy Comparison

The Clean Slate

Dropping and recreating the log ensures a perfect match but discards all pending changes, making a full refresh mandatory.

The In-Place Evolution

Using ALTER commands to update the log preserves existing data, maintaining incremental refresh capabilities.

Method 1: The Clean Slate

-  **Hard Reset:** Wipes all metadata and MLOG\$ table content completely.
-  **Pending Changes:** Any DML captured before the drop is permanently lost.
-  **Impact:** Forces a resource-heavy COMPLETE refresh for all dependent MVs.

Method 2: In-Place Evolution

The `ALTER MV LOG ADD` command updates metadata without clearing logs. This is the preferred path for production systems.

It ensures continuous FAST refresh capability and avoids the heavy performance cost of a complete table scan during replication.



Decision Matrix

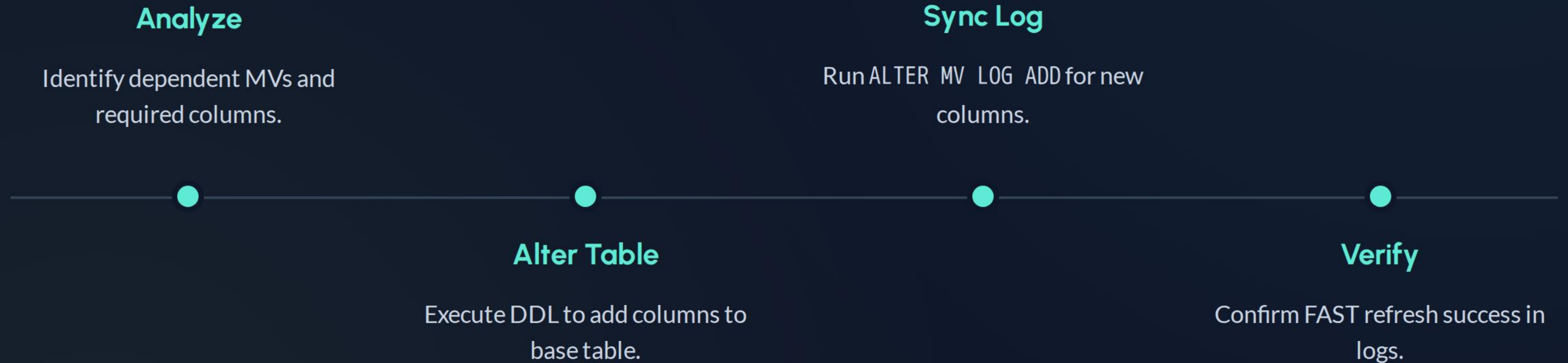
Metric	Clean Slate (Recreate)	In-Place (Alter)
Data Loss	High (Pending rows lost)	None (Changes preserved)
MV Sync	Forces Complete Refresh	Maintains Fast Refresh
Maintenance	High (Locks source table)	Low (Metadata update)

Refresh Performance Analysis



**Simulation based on 100GB source table with 1% daily change rate.*

Implementation Workflow

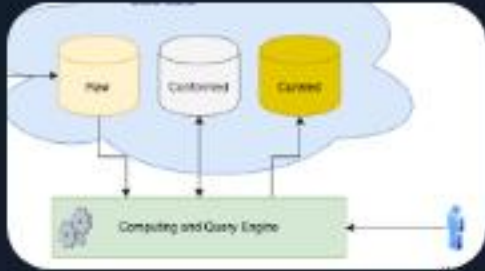


Questions?

Ensuring your replication pipeline remains robust.

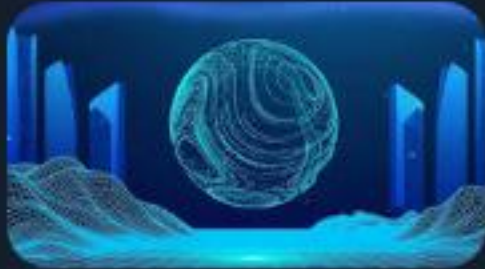
Database Administration & Architecture Team

Image Sources



https://miro.medium.com/v2/resize:fit:666/1*vh1hFhWU0w0CzCiA64Xy3A.png

Source: medium.com



https://static.vecteezy.com/system/resources/previews/066/025/500/non_2x/city-building-smart-background-technology-future-line-abstract-data-business-lights-skyline-network-concept-blue-office-3d-architecture-industry-innovation-landscape-futuristic-neon-cyber-vector.jpg

Source: www.vecteezy.com